

Microalgae for the Treatment of Wastewater and Biofuel

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Motivation/Background

- **Motivation:** Grow microalgae as a 3rd generation fuel source while providing a secondary service of wastewater treatment
- **Background:** The need to find renewable fuel sources is growing and microalgae is being looked at as a 3rd generation fuel source. Microalgae is typically grown using synthetic fertilizer, meanwhile nitrate and phosphate are particularly expensive to remove from wastewater and can negatively impact water quality.
- **The ESIP IRES program**
 - Emphasis on self sufficient means for indigenous communities to cultivate energy
 - Promotes students learning intercultural research skills in a hands-on environment abroad

Methods

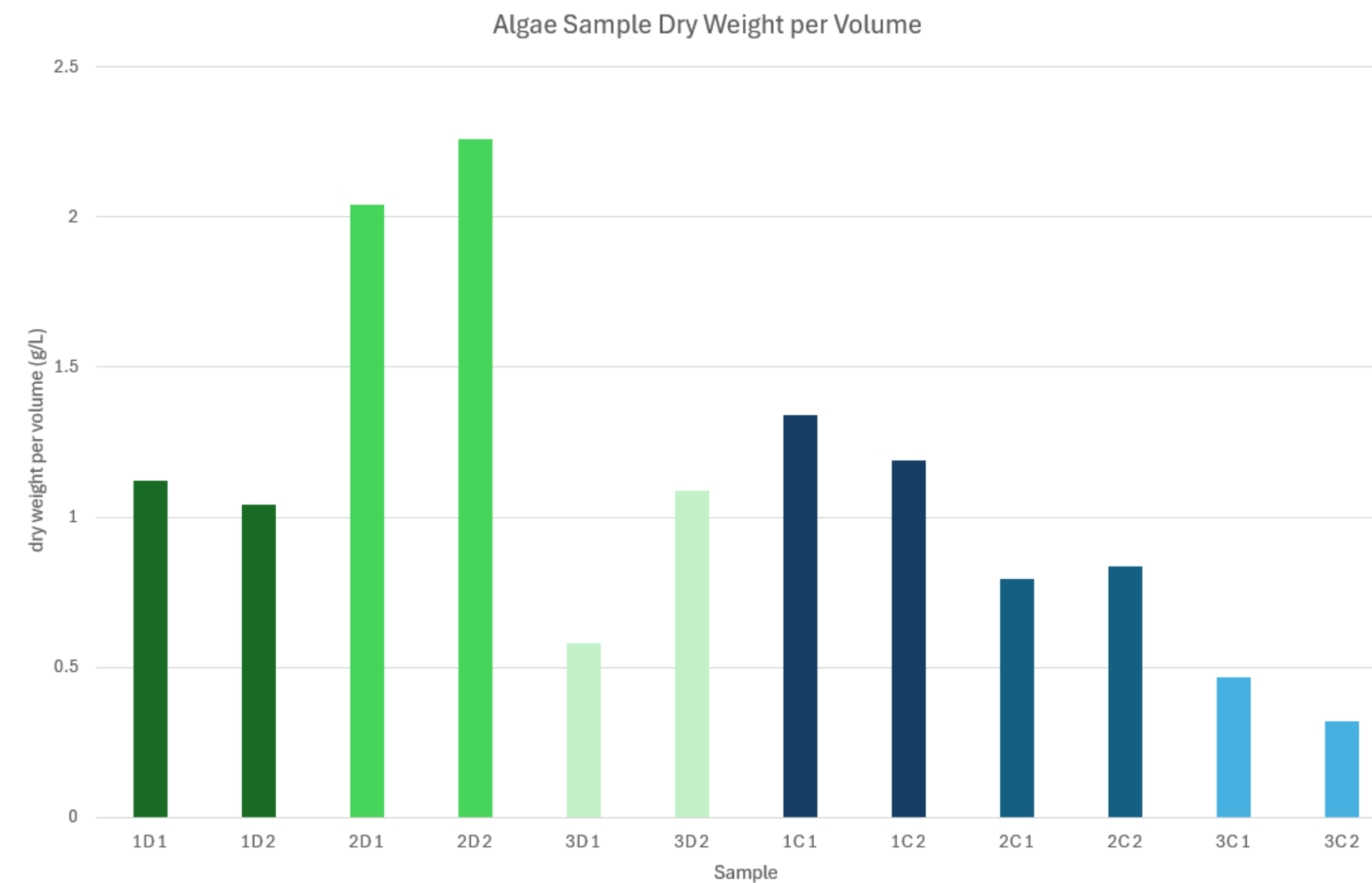
- Two strains of algae were cultivated at 3 different nitrate and phosphate concentrations each, Each strain was fed carbon dioxide and agitated through pvc lines connected to an air pump and carbon dioxide cannister.
- UV Spectrometry was used to measure Nitrate and Phosphate Concentrations daily
- The Dry weight was taken to determine which strain grew the most under the different sets of conditions

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** Pictures were taken left to right at Hallet Cove and Adelaide Zoo

Results



- **Highest biomass per volume:** Sample 2D
 - For areas that have higher phosphate runoff from fertilizer *desmodesmus* would be the better option for algal strain
- **Lowest biomass per volume:** Sample 3C

Discussion

- Creating a growth media similar to regional agricultural runoff since it varies regionally still needs to be done
- Extraction and measurement of lipids from algae grown on similar conditions also still needs to be completed

Conclusions

- *Desmodesmus* grew faster and denser than *Chlorella Vulgaris* (control, high phosphate, high nitrate).
- Phosphate and nitrate conditions affect growth for both strains
- The ESIP IRES program has increased my understanding of the research process and helped me cultivate skills that are essential for working in diverse intercultural environments

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WHERE DISCOVERIES BEGIN

